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A 4.21-to-15.18 GHz Magnetically Reconfigurable and Fully Symmetrical Quad-Core Quad-Mode VCO

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***Abstract*— This work proposes a wideband low phase noise (PN) quad-core quad-mode voltage-controlled oscillator (VCO) which features magnetically reconfigurable (MR) technique and fully symmetrical architecture. The MR technique, realized through a transformer-based tank integrating an embedded switched inductor, leverages varying core excitation currents to achieve four distinct equivalent inductances, thereby enabling quad-mode operation. This approach avoids the frequency tuning range (FTR) compromise encountered when using capacitive tuning elements in multi-mode tanks. Furthermore, the fully symmetrical characteristics of the inductive, capacitive, and active elements, along with the routing layout, minimize mismatches among the VCO cores, ensuring effective synchronization of the four single-core VCOs and optimizing the PN in quad-core operation. The proposed quad-core quad-mode VCO achieves an FTR of 113.2% (4.21 to 15.18 GHz), a PN of −141.4 dBc/Hz at a 10 MHz offset from 4.28 GHz, a peak figure-of-merit (FoM) of 190.5 dBc/Hz, a peak FoM_T of 211.6 dBc/Hz and a peak FoM_{TA} of 220.5 dBc/Hz at a 10 MHz offset.**

***Index Terms*— Magnetically reconfigurable (MR), phase noise (PN), quad-core, quad-mode, voltage-controlled oscillator (VCO), wide frequency tuning range (FTR).**

I. Introduction

TO MEET the requirements on multi-standard and multi-application scenarios in 5G communications, a local oscillator (LO) with wide frequency tuning range (FTR) is urgently desired. Besides, with the increasing demand for signal purity in high-order modulation schemes such as quadrature amplitude modulation (QAM), voltage-controlled oscillators (VCOs) are required to maintain low phase noise (PN). However, achieving wide bandwidth and low PN simultaneously remains a significant challenge for VCO designs.

Compared to the scheme that uses multiple individual VCOs for multi-band operation, mode-switching technique summarized in [1] provides a more efficient solution, enabling multi-band coverage with reduced area and lower power consumption. The switched-inductors or switched-transformers techniques are favorable to achieve wide bandwidth by utilizing multiple effective inductance values [2], [3], [4]. Although these techniques can broaden the bandwidth effectively in a simple manner, the ON-resistance of the switches degrades the Q factor of the tank, thus significantly deteriorating the PN. Recently, coupling multiple VCO cores operating at different modes via mode switches has been demonstrated to be a promising method to simultaneously achieve low PN and wide FTR.

Based on Leeson's PN model [5], the PN can be expressed as follows:

$$L(\Delta\omega) = 10\log\left[\frac{FkTR_T}{V_0^2 Q^2}\left(\frac{\omega_0}{\Delta\omega}\right)^2\right] \tag{1}$$

where k is Boltzmann's constant, T is the absolute temperature, F is the device excess noise factor, V_0 is the oscillation amplitude, R_T is the tank resistance, and Q is the tank quality factor. Coupling N VCO cores results in a $10\log N$ dB PN reduction, as R_T in (1) is reduced by a factor of N while other parameters remain fixed. However, this PN performance improvement comes at the cost of an N-fold increase in power consumption. By utilizing mode switches among the multiple VCO cores, multiple operation modes can be generated, extending the FTR without incurring switch losses.

Dual-mode VCOs have been widely reported [6], [7], [8], [9], [10], [11], [12], [13], which employ either magnetically reconfigurable (MR) inductance (denoted "M") or electric coupling via added capacitors to change the effective capacitance (denoted "E") to realize dual-mode operation. While an octave tuning range can theoretically be divided or multiplied to target any frequency band, dual-mode VCOs typically struggle to achieve full octave coverage due to practical limitations. As a result, triple-mode [14], [15], [16], [17] or even quad-mode VCOs [18], [19], [20], [21], [22], [23], [24] have

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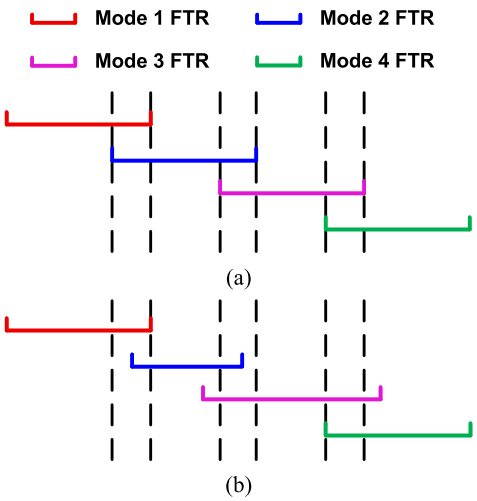


Fig. 1. (a) Uniform FTR in each mode. (b) Nonuniform FTR in each mode.

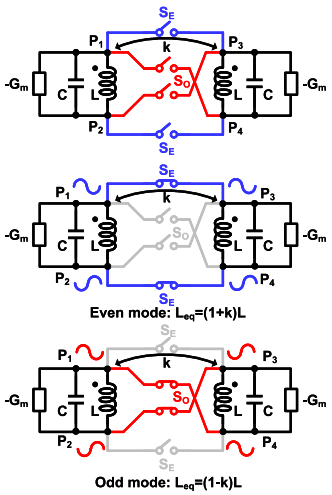


Fig. 2. Operating principle of dual-core dual-mode VCO.

gained popularity. In order to expand the number of operation modes, MR technique is typically combined with electric coupling. However, the latter introduces substantial parasitic capacitance, which increases $C_{\min}$ and consequently reduces the FTR, expressed as follows:

$$\mathrm{FTR} = 2 \times \frac{\sqrt{C_{\max}/C_{\min}} - 1}{\sqrt{C_{\max}/C_{\min}} + 1} \tag{2}$$

where $C_{\min}$ and $C_{\max}$ denote the minimum and maximum values of the tank capacitance, respectively. Thus, MR technique is favored in those multi-core multi-mode VCO designs.

In multi-core multi-mode VCOs, transformer-based resonant tanks are preferred for their design flexibility and area efficiency. However, as the number of operation modes increases, transformer design becomes challenging and the coupling network becomes complex, both of which tend to introduce asymmetries across VCO cores. As a result, the asymmetry induced by transformers or even other components inevitably introduces frequency mismatches in multi-core operation, which theoretically degrade the PN performance. Therefore, a fully symmetrical architecture in multi-core multi-mode VCOs are preferred to eliminate the negative impact on the PN performance.

A substantial number of state-of-the-art triple-mode or quad-mode VCOs have been proposed in recent years. A dual-core triple-mode VCO with a wide FTR of 80.6% and a PN of -137.9 dBc/Hz at a 10 MHz offset from 7.15 GHz is proposed by using Electrical-Magnetic (E-M) mixed coupling in [15]. However, the asymmetrical inductance seen from each core causes inter-core mismatches that degrades the PN. Moreover, the inclusion of additional mode switches and switched capacitor arrays (SCAs) impose extra parasitics, thus reducing the FTR. A quad-core quad-mode VCO in [22] achieves an FTR of 101% and a PN of -133.2 dBc/Hz at a 10 MHz offset from 13.7 GHz by using E-M mixed coupling, whereas the issue of asymmetrical inductance persists, and the parasitics from electric coupling capacitors negatively impact the FTR. Additionally, the non-uniformity of the FTR across each mode is also undesirable. With the same total FTR, a uniform FTR in each mode yields roughly equal overlaps between adjacent modes, as shown in Fig. 1(a). In contrast, a nonuniform FTR makes the overlaps highly uneven, as shown in Fig. 1(b). This gives rise to two concerns. First, large differences in overlap make the band-decision thresholds difficult to unify and require additional algorithm adaptation, which increases system complexity. Second, under PVT variations, the smaller overlap is more likely to shrink and create a frequency gap between adjacent modes. Other works still suffer from these issues [18], [21]. In summary, although multi-core multi-mode VCOs have demonstrated superior performance in terms of FTR and PN, issues such as asymmetrical inductance, parasitic-induced FTR reduction, and nonuniform FTR still persist, ultimately degrading the overall performance of the VCOs.

To surmount these issues, this work proposes a quad-core quad-mode VCO that achieves ultra-wide bandwidth and low PN by employing MR technique and fully symmetric architecture, which effectively eliminates the asymmetry and extra parasitics, and ensures a uniform FTR across all modes. This article is organized as follows. Section II presents the structure of the proposed VCO and the operating principles of the four modes. Section III discusses the symmetry of the design and its optimization of performance. Section IV presents the measurement results and Section V draws the conclusion.

II. Proposed Quad-Mode Operation Utilizing MR Technique

A. Triple-Mode Operating Principle

Dual-core dual-mode VCO serves as the foundational architecture for multi-core multi-mode VCO designs, as shown in Fig. 2. By controlling the phases of the two coupled inductors through the mode switches S_O and S_E, two effective inductances are achieved, thereby realizing two operation modes, namely odd mode and even mode. Fig. 3 shows the quad-core triple-mode configuration derived from the dual-core dual-mode structure. It simply connects four single VCO cores through mode switches, with the key difference that

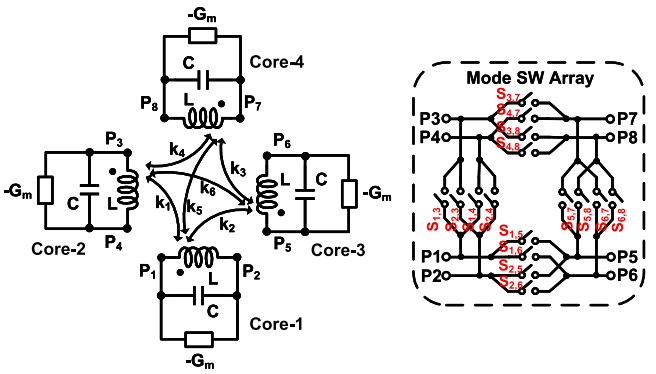


Fig. 3. Configuration of the quad-core triple-mode VCO.

the inductors of the four cores are mutually coupled with each other. Besides, to ensure symmetry among the cores, the coupling coefficients between adjacent inductors and those between the opposite inductors are equal, resulting in the relationship as follows:

$$k_1 = k_2 = k_3 = k_4 = k_{\text{adj}} \tag{3}$$

$$k_5 = k_6 = k_{\text{opp}}. \tag{4}$$

The sixteen mode switches ($S_{i,j}$) are employed to connect the corresponding two ports (P_i, P_j). By changing the current directions of the tank inductors in each core through the mode switches, the proposed VCO can operate in three resonant modes. The three operation modes are described as follows.

1) In mode-L, as shown in Fig. 4(a), the current directions of four tank inductors are all in-phase, with the switches $S_{1,3}$, $S_{1,5}$, $S_{2,4}$, $S_{2,6}$, $S_{3,7}$, $S_{4,8}$, $S_{5,7}$, and $S_{6,8}$ turned on. Thus, the voltages at ports $P_{1,3,5,7}$ are in phase, as are those at $P_{2,4,6,8}$. The equivalent inductance of each core is expressed as follows:

$$L_{\text{mode},L} = \left(1 + k_{\text{adj}} + k_{\text{adj}} + k_{\text{opp}}\right)L = L + 2M_{\text{adj}} + M_{\text{opp}} \tag{5}$$

where $M_{\text{adj}} = k_{\text{adj}}L$ and $M_{\text{opp}} = k_{\text{opp}}L$.

2) In mode-M, as illustrated in Fig. 4(b), the currents in the adjacent cores of any given core are either in-phase or anti-phase, whereas the current in the opposite core is strictly anti-phase. In this mode, switches $S_{1,4}$, $S_{1,5}$, $S_{2,3}$, $S_{2,6}$, $S_{3,7}$, $S_{4,8}$, $S_{5,8}$, and $S_{6,7}$ are turned on, leading to ports $P_{1,4,5,8}$ (as well as $P_{2,3,6,7}$) connected together. Specifically, for core-1, the magnetic influence from core-2 and -3 cancels out, and only core-4 contributes a mutual inductance of $-M_{\text{opp}}$. Thus, the equivalent inductance of core-1 is expressed as follows:

$$L_{\text{mode},M} = \left(1 - k_{\text{adj}} + k_{\text{adj}} - k_{\text{opp}}\right)L = L - M_{\text{opp}}. \tag{6}$$

Apart from core-1, it can be observed that the equivalent inductances for other three cores are also equal to $L_{\text{mode},M}$. Additionally, since $L_{\text{mode},L}$ is larger than $L_{\text{mode},M}$, the operating frequency of mode-L is lower.

3) In mode-H, as shown in Fig. 4(c), the currents in adjacent cores are anti-phase, whereas those in opposite cores are in-phase, with the switches $S_{1,4}$, $S_{1,6}$, $S_{2,3}$, $S_{2,5}$, $S_{3,8}$, $S_{4,7}$, $S_{5,8}$, and $S_{6,7}$ turned on. Each core experiences mutual inductances of $-2M_{\text{adj}}$ from its two adjacent cores and M_{opp} from the opposite core, resulting in the equivalent inductance seen from each core being

$$L_{\text{mode},H} = \left(1 - 2k_{\text{adj}} + k_{\text{opp}}\right)L = L - 2M_{\text{adj}} + M_{\text{opp}}. \tag{7}$$

Since $L_{\text{mode},M} - L_{\text{mode},H}$ is equal to $2(M_{\text{adj}} - M_{\text{opp}})$ and M_{adj} is set larger than M_{opp} in this design, mode-M is thus operating at a lower frequency.

4) In mode-N, as shown in Fig. 4(d), the current in any given core is anti-phase with other three cores, with the switches $S_{1,4}$, $S_{1,6}$, $S_{2,3}$, $S_{2,5}$, $S_{3,8}$, $S_{4,7}$, $S_{5,8}$, and $S_{6,7}$ turned on. Based on the current direction in each core, the effective inductance of each core can be derived as $L + M_{\text{opp}}$ for core-1 and -4, $L + 2M_{\text{adj}} - M_{\text{opp}}$ for core-2, and $L - 2M_{\text{adj}} - M_{\text{opp}}$ for core-3. Unlike the other three modes, where the effective inductor for each core is identical, the inductance seen from each core in this mode is varied, leading to significant frequency mismatches among four cores. In a multi-core VCO, all cores should be typically designed to be identical to ensure balanced operation. This mismatch in inductance will lead to unequal current distribution among the cores, which in turn causes imbalances in the waveforms among VCO cores. Then the current will flowthrough the lossy switches in order to equalize the energy of the resonators, thereby forcing the VCO to oscillate at the same frequency. This process, however, degrades the tank Q and renders the theoretical 6 dB PN optimization from the quad-core operation unachievable. Therefore, this mode is abandoned.

In summary, three distinct equivalent inductance values are realized through a fully symmetrical transformer network composed of four mutually coupled inductors. However, since the difference $L_{\text{mode},L} - L_{\text{mode},M}$ exceeds $L_{\text{mode},M} - L_{\text{mode},H}$ by $4M_{\text{opp}}$, achieving continuous frequency tuning between mode-M and -H may result in a frequency gap between mode-L and -M. Therefore, an additional mode inserted between mode-L and -M is required, which should minimally affect other modes.

B. Quad-Mode Operating Principle

Based on the triple-mode transformer in Fig. 3, a switched inductor is embedded within the four tank inductors, as shown in Fig. 5, thereby creating one additional mode. The switched inductor operates in two different states, depending on its connection state.

1) 1) When the switched inductor L_{sw} is turned off, three distinct modes-mode-1, -3, and -4, as shown in Fig. 6(a), (c), and (d), operate following the same principle as the original triple-mode configuration, corresponding to mode-L, -M, and -H, respectively.
2) When L_{sw} is turned on, the induced current in L_{sw} is canceled in mode-M and -H by two opposing currents from the four tank inductors. Therefore, turning on the switch S_{Ls} does not affect the operating frequencies of

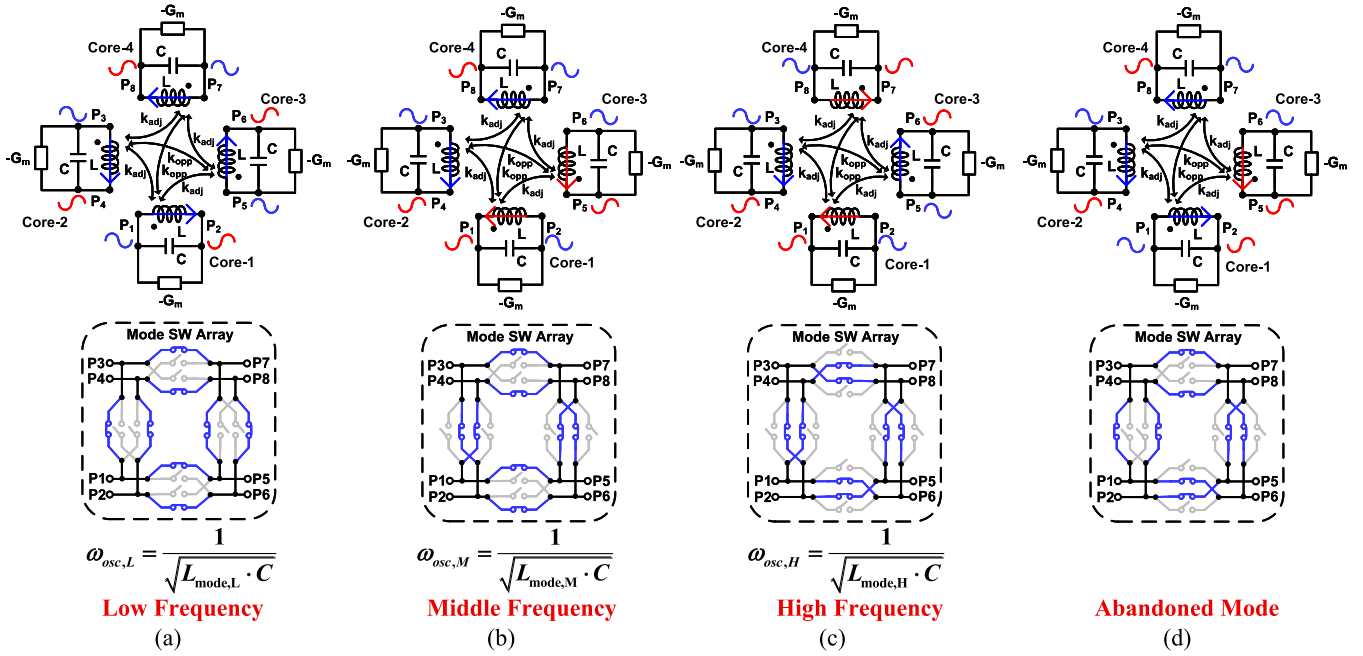


Fig. 4. Operating principle of the quad-core triple-mode VCO. (a) Mode-L. (b) Mode-M. (c) Mode-H. (d) Mode-N.

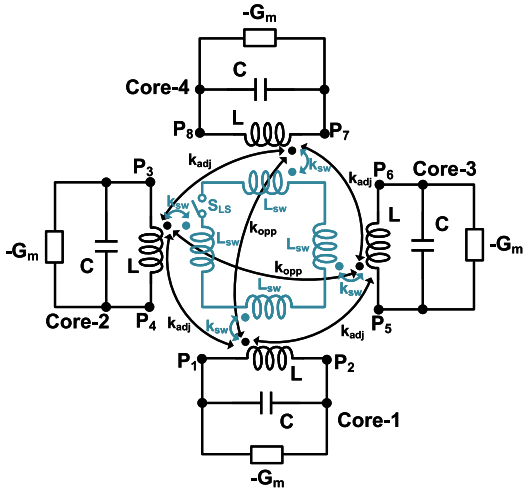


Fig. 5. Structure of the proposed quad-core quad-mode VCO.

mode-M and -H. In mode-L, however, the currents in the four tank inductors are aligned in the same direction, so the induced current in L_{sw} cannot be canceled. Consequently, an additional operation mode, denoted as mode-2, is inserted between mode-1 and -3, which is implemented by controlling L_{sw} through the mode-L configuration, as shown in Fig. 6(b).

Since the switched inductor is controlled by an NMOS switch (S_{Ls}), it should be noted that due to the non-ideal characteristics of the switch, it can be equivalent to an ON-resistance (r_{ON}) when turned on, and as two parasitic capacitances (C_{OFF}) to ground when turned off, as shown in Fig. 7. The presence of r_{ON} and C_{OFF} will affect the equivalent inductance in each operation mode, which can be analyzed using a simplified lumped model of the switched-inductor-based quad-mode transformer, as shown in Fig. 8.

1) When the switch S_{Ls} is turned on corresponding to mode-2, the switched inductor is connected in series with the ON-resistance r_{ON}, as shown in Fig. 8(a). The electrical parameters–namely k_{adj}, k_{opp} and L–will be affected by r_{ON}, resulting in modified values denoted as $k_{adj,ON}$, $k_{opp,ON}$ and L_{on}, respectively. The modified parameter $k_{adj,ON}$ can be derived based on Fig. 8(a) by analyzing $Z_{12} = V_1/I_2$ under the condition $I_1 = I_3 = I_4 = 0$, while I_{sw} and I_2 provide current excitations. According to the mutual inductance voltage equation [25], the voltage V_1 can be written as follows:

$$V_1 = sM_{adj}I_2 + sM_{sw}I_{sw} \tag{8}$$

where $M_{sw} = k_{sw}(LL_{sw})^{1/2}$ and $s = jw$.
As the ON-resistance r_{ON} effectively connects both ends of the switched inductor, the Kirchhoff's voltage law (KVL) equation can be derived as follows:

$$0 = sL_{sw}I_{sw} + sM_{sw}I_2 + r_{ON}I_{sw}. \tag{9}$$

From (8) and (9), Z_{12} can be derived as follows:

$$Z_{12} = sM_{adj,ON} = s\left(M_{adj} - \frac{sM_{sw}^2}{sL_{sw} + r_{ON}}\right). \tag{10}$$

It can be seen from (10) that the presence of r_{ON} leads to variation in the mutual inductance M_{adj} between adjacent cores, where the modified value $M_{adj,ON}$ can be expressed as follows:

$$M_{adj,ON} = M_{adj} - \frac{sM_{sw}^2}{sL_{sw} + r_{ON}}. \tag{11}$$

Similarly, $M_{opp,ON}$ and L_{ON} can be derived as follows:

$$M_{opp,ON} = M_{opp} - \frac{sM_{sw}^2}{sL_{sw} + r_{ON}} \tag{12}$$

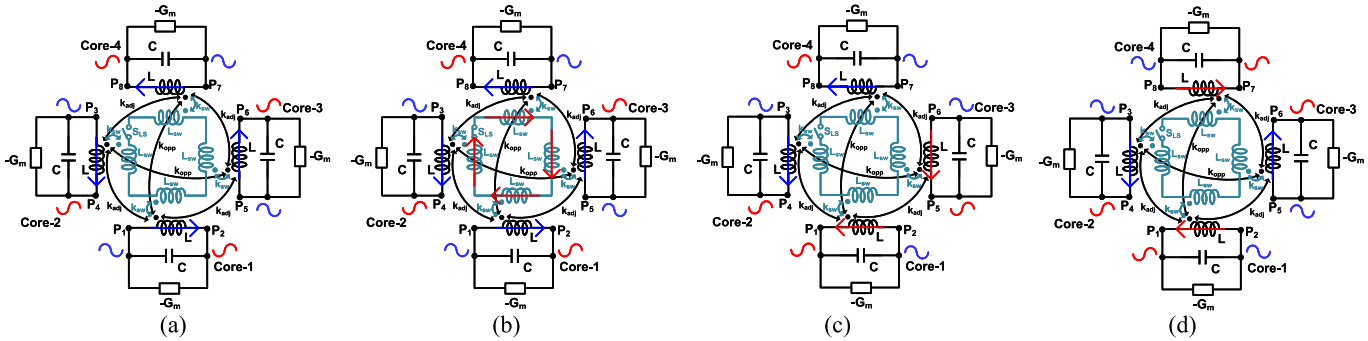


Fig. 6. Operating principle of the proposed quad-core quad-mode VCO. (a) Mode-1. (b) Mode-2. (c) Mode-3. (d) Mode-4.

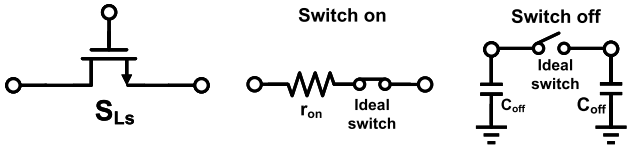


Fig. 7. NMOS switch of the switched inductor and its equivalent circuits in ON and OFF states.

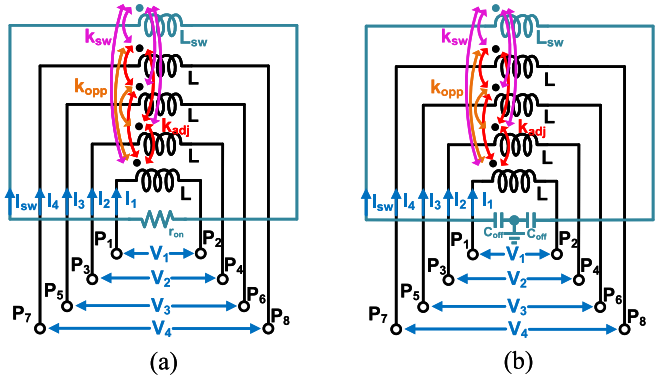


Fig. 8. Simplified lumped model of the proposed transformer with the switch S_{Ls} in (a) ON and (b) OFF states.

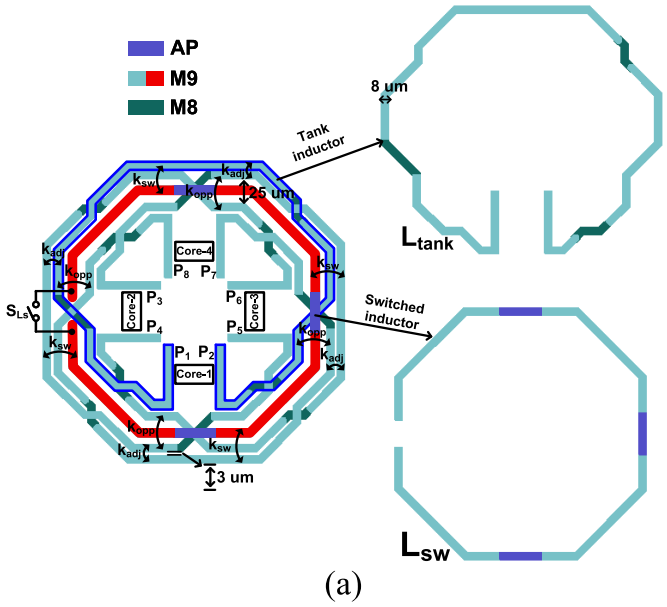
$$L_{\mathrm{ON}} = L - \frac{sM_{\mathrm{sw}}^2}{sL_{\mathrm{sw}} + r_{\mathrm{ON}}}. \tag{13}$$

Consequently, the equivalent inductance in mode-2 can be derived as follows:

$$L_{\mathrm{mode,2}} = L + 2M_{\mathrm{adj}} + M_{\mathrm{opp}} - \frac{4sM_{\mathrm{sw}}^2}{sL_{\mathrm{sw}} + r_{\mathrm{ON}}}. \tag{14}$$

As long as the size of the switch S_{Ls} is sufficiently large, its ON-resistance r_{ON} can be neglected and the difference in equivalent inductances between mode-1 and -2 becomes $4k_{\mathrm{sw}}^2 L$. Therefore, by adjusting the spacing between L and L_{sw}, the coupling coefficient k_{sw} can be controlled, enabling the separation of $L_{\mathrm{mode,2}}$ from $L_{\mathrm{mode,1}}$, and allowing mode-2 to fill the frequency gap between mode-1 and -3.

2) When the switch S_{Ls} is turned off corresponding to mode-1, -3, and -4, the switched inductor is connected in series with the parasitic capacitance C_{OFF}, as shown in Fig. 8(b). Similar to the previous analysis, the only difference lies in replacing r_{ON} with $2/(sC_{\mathrm{OFF}})$. Thus, the equivalent modified $M_{\mathrm{adj,OFF}}$, $M_{\mathrm{opp,OFF}}$ and L_{OFF} can be expressed as follows:



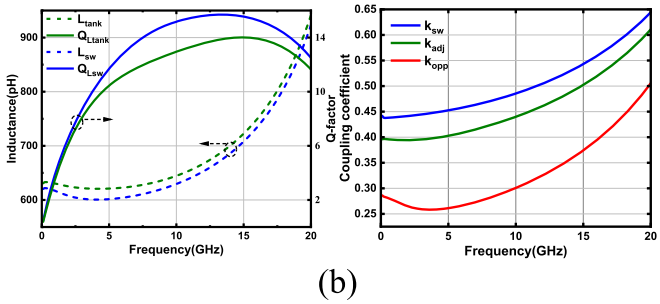


Fig. 9. (a) Transformer structure of the proposed quad-core quad-mode VCO. (b) Its simulated parameters obtained from EM simulations.

$$M_{\mathrm{adj,OFF}} = M_{\mathrm{adj}} - \frac{s^2 M_{\mathrm{sw}}^2 C_{\mathrm{OFF}}}{s^2 L_{\mathrm{sw}} C_{\mathrm{OFF}} + 2} \tag{15}$$

$$M_{\mathrm{opp,OFF}} = M_{\mathrm{opp}} - \frac{s^2 M_{\mathrm{sw}}^2 C_{\mathrm{OFF}}}{s^2 L_{\mathrm{sw}} C_{\mathrm{OFF}} + 2} \tag{16}$$

$$L_{\mathrm{OFF}} = L - \frac{s^2 M_{\mathrm{sw}}^2 C_{\mathrm{OFF}}}{s^2 L_{\mathrm{sw}} C_{\mathrm{OFF}} + 2}. \tag{17}$$

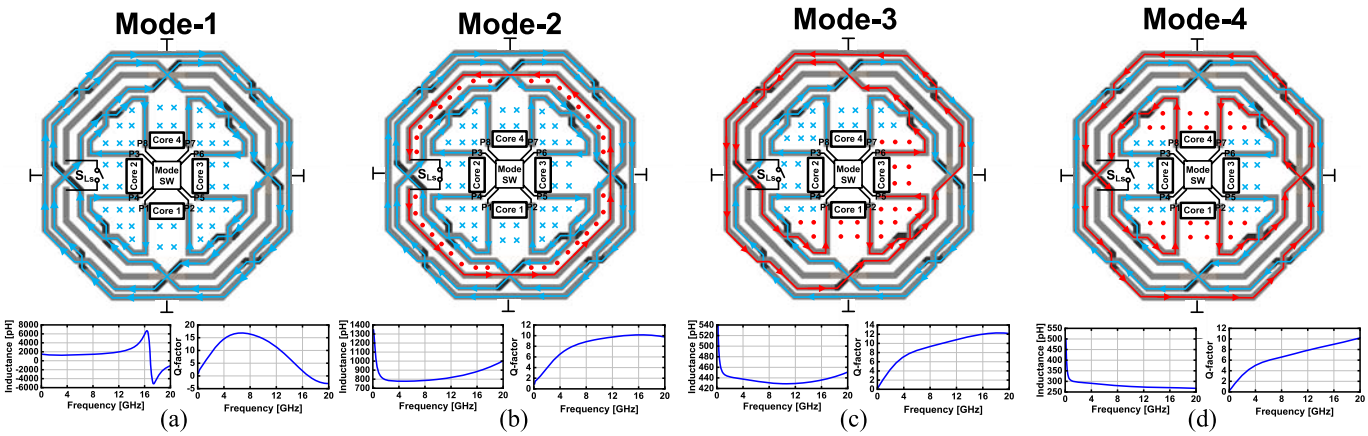


Fig. 10. Coil currents, magnetic field distribution and simulated effective L and Q in (a) mode-1, (b) mode-2, (c) mode-3 and (d) mode-4.

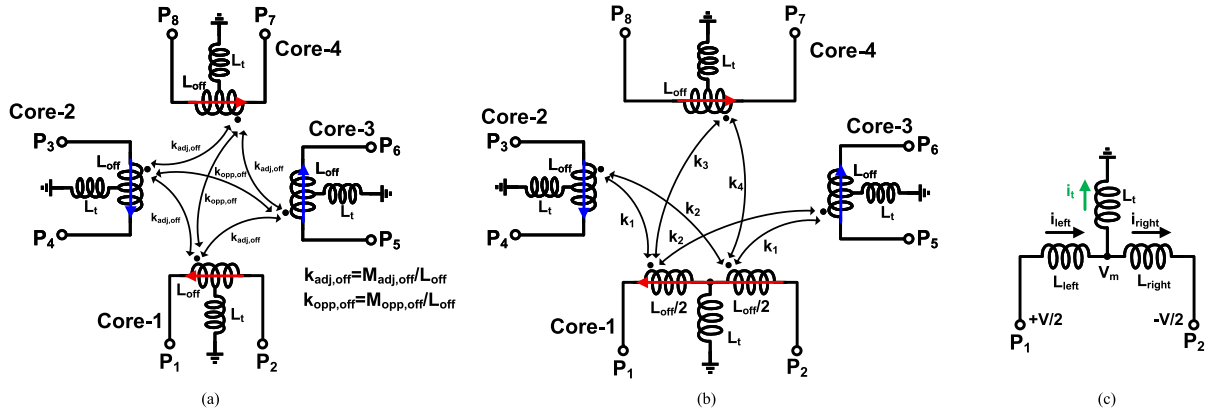


Fig. 11. Circuit equivalence for the mode-3 transformer with the switched inductor off.

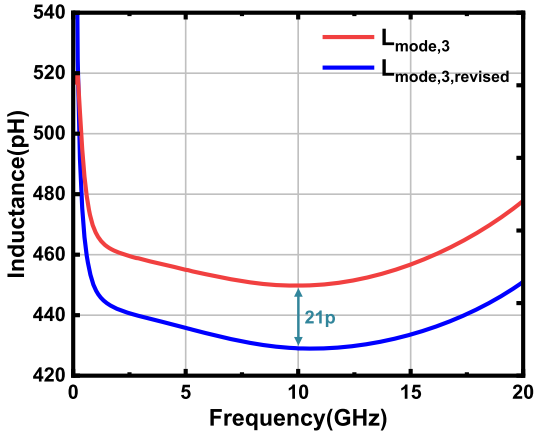


Fig. 12. Comparison of mode-3 effective inductance: original versus revised.

As a result, the equivalent inductances in mode-1, -3, and -4 can be expressed as follows:

$$L_{mode,1} = L + 2M_{adj} + M_{opp} - \frac{4s^2 M_{sw}^2 C_{OFF}}{s^2 L_{sw} C_{OFF} + 2} \approx L + 2M_{adj} + M_{opp} \quad (18)$$

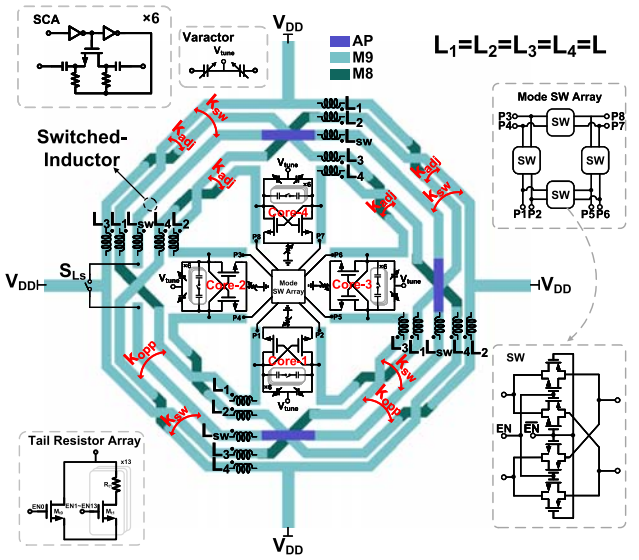
$$L_{mode,3} = L_{OFF} - M_{opp,OFF} = L - M_{opp} \quad (19)$$

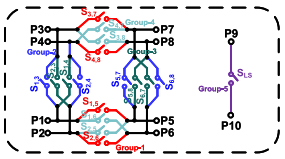
$$L_{mode,4} = L_{OFF} - 2M_{adj,OFF} + M_{opp,OFF} = L - 2M_{adj} + M_{opp}. \quad (20)$$

In (18), the last term is a minor factor and can be neglected. Therefore, the equivalent inductance in mode-1 remains approximately unchanged compared to (5). In mode-3 and -4, the equivalent inductances remain consistent with (6) and (7), respectively, as indicated in (19) and (20). In other words, the equivalent inductances in mode-1, -3, and -4 are rarely affected by the switched inductor. In mode-1, the result comes from an expression approximation, while in mode-3 and -4, it can be explained by the fact that the currents induced in L_{sw} by the surrounding tank inductors cancel each other out, which means that L_{sw} is transparent in mode-1, -3, and -4.

C. Proposed Transformer Tank

Fig. 9(a) shows the layout of the proposed transformer, which consists of four identical single-coil tank inductors (L_{tank}) and an extra switched inductor (L_{sw}), intertwined in a center-symmetric manner. The coupling coefficients are identical among any pairs of adjacent inductors, any pairs of opposite inductors, and between each tank inductor and the switched inductor, denoted as k_{adj}, k_{opp}, and k_{sw}, respectively. To mitigate the area overhead of the multi-core structure, the





	Group-1 Bit-1				Group-2 Bit-2				Group-3 Bit-3				Group-4 Bit-4				Group-5 Bit-5
SW	$S_{1,5}$	$S_{2,6}$	$S_{3,7}$	$S_{4,8}$	$S_{1,3}$	$S_{2,4}$	$S_{5,7}$	$S_{6,8}$	$S_{1,4}$	$S_{2,3}$	$S_{5,8}$	$S_{6,7}$	$S_{1,6}$	$S_{2,5}$	$S_{3,8}$	$S_{4,7}$	S_{Ls}
Mode-1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
Mode-2	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1
Mode-3	1	1	1	1	0	0	0	0	1	1	1	1	0	0	0	0	0
Mode-4	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0

(b)

Fig. 13. (a) Detailed implementation of the proposed quad-core quad-mode VCO. (b) Mode configuration truth table (*1=switch on*, *0=switch off*).

transformer efficiently concentrates the magnetic flux within the inductor coils, maximizing flux utilization. The switched inductor is placed in the middle of the tank inductors and tightly coupled with other coils. Both the tank inductors and the switched inductor have the width of 8 μm. Additionally, to ensure that $k_{\text{adj}} > k_{\text{opp}}$, the coil spacing between adjacent tank inductors is set to 3 μm, while the spacing between opposite tank inductors is set as 25 μm. As shown in Fig. 9(b), EM simulation results show that the self-inductance and Q factor of the tank inductors are 645 pH and 13 at 10 GHz, respectively, while those of the switched inductor are 630 pH and 15. The adjacent and opposite coupling coefficients, k_{adj} and k_{opp}, are found to be 0.44 and 0.3 at 10 GHz, respectively. Besides, the coupling coefficient k_{sw} between the switched inductor and the tank inductors is 0.49 at 10 GHz.

Fig. 10 illustrates the magnetic flux of the proposed quad-mode operation, along with the simulated equivalent inductance and equivalent Q factor. Fig. 10(a) and (b) corresponds to the cases of mode-1 and -2, where the magnetic fluxes among all tank inductors are mutually reinforced. In mode-1, the switched inductor L_{sw} is disconnected and the equivalent inductance is equal to (18). In mode-2, when S_{Ls} is shorted, an induced current flowing through L_{sw} generates opposite magnetic flux. The equivalent inductance is thus equated to (14). In Fig. 10(c) and (d), corresponding to mode-3 and -4, the magnetic fluxes in any two cores are anti-phased with those in the other two cores. Mode-4 follows (20). For mode-3, (19) becomes inaccurate since magnetic coupling breaks the left–right symmetry of the spiral halves. Consequently, the differential analysis needs to include the tap inductance, and the expression for $L_{\text{mode},3}$ should be updated accordingly.

Fig. 11(a) shows the quad-mode transformer for mode-3 with the switched inductor off, where $k_{\text{adj,OFF}} = M_{\text{adj,OFF}}/L_{\text{OFF}}$ and $k_{\text{opp,OFF}} = M_{\text{opp,OFF}}/L_{\text{OFF}}$. From the perspective of core-1, we split the inductor L_{OFF} at its geometric center into two equal halves and retain only the couplings relevant to core-1, as shown in Fig. 11(b). To preserve the total mutual coupling for core-1, the coefficients satisfy $k_1 + k_2 = \sqrt{2}k_{\text{adj,OFF}}$ and $k_3 + k_4 = \sqrt{2}k_{\text{opp,OFF}}$. Further inspection of the transformer layout in Fig. 9(a) shows that $k_1 < k_2$ and $k_3 = k_4$. Under these conditions, the equivalent inductances of the left and right halves are obtained as follows:

$$L_{\text{left}} = \frac{L_{\text{OFF}}}{2} - k_1\sqrt{L_{\text{OFF}} \cdot \frac{L_{\text{OFF}}}{2}} + k_2\sqrt{L_{\text{OFF}} \cdot \frac{L_{\text{OFF}}}{2}} - \frac{1}{2}M_{\text{opp,OFF}} \tag{21}$$

$$L_{\text{right}} = \frac{L_{\text{OFF}}}{2} - k_2\sqrt{L_{\text{OFF}} \cdot \frac{L_{\text{OFF}}}{2}} + k_1\sqrt{L_{\text{OFF}} \cdot \frac{L_{\text{OFF}}}{2}} - \frac{1}{2}M_{\text{opp,OFF}}. \tag{22}$$

Since $L_{\text{left}} \neq L_{\text{right}}$, a current ($i_t$) flows into the tap inductor, as shown in Fig. 11(c). Applying a differential excitation of $+V/2$ at P_1 and $-V/2$ at P_2, the effective inductance of mode-3 is obtained as follows:

$$L_{\text{mode},3,\text{revised}} = \frac{V}{\frac{1}{2}\left(i_{\text{left}} + i_{\text{right}}\right)} = \frac{4\left(L_{\text{left}}L_{\text{right}} + L_{\text{left}}L_t + L_{\text{right}}L_t\right)}{L_1 + L_2 + 4L_t}. \tag{23}$$

In the limiting cases, the results are intuitive. When the tap path is open ($L_t \to \infty$), the effective inductance becomes $L_{\text{left}} + L_{\text{right}}$. When the tap is ideally grounded ($L_t \to 0$), $L_{\text{mode},3,\text{revised}}$ approaches $4L_{\text{left}}L_{\text{right}}/(L_{\text{left}} + L_{\text{right}})$. For the symmetric case $L_{\text{left}} = L_{\text{right}}$, we always have $L_{\text{mode},3,\text{revised}} = 2L_{\text{left}} = 2L_{\text{right}}$ independent of L_t.

Owing to the tap inductor, mode-3 shows a smaller effective inductance. The revised effective inductance of mode-3 is 21 pH lower than the original, as shown in Fig. 12.

The simulated equivalent inductance and Q factor are 1298 pH and 16 at 5 GHz in mode-1, 782 pH and 8.5 at 7 GHz in mode-2, 429 pH and 10.1 at 10 GHz in mode-3, and 272 pH and 8.2 at 13 GHz in mode-4, as shown in Fig. 10. It can be observed that the equivalent inductance decreases progressively from mode-1 to -4.

III. Design and Analysis of The Proposed Quad-Core Quad-Mode VCO

A. Proposed VCO Design

Fig. 13(a) shows the schematic of the proposed quad-core quad-mode VCO, which consists of four identical single-core

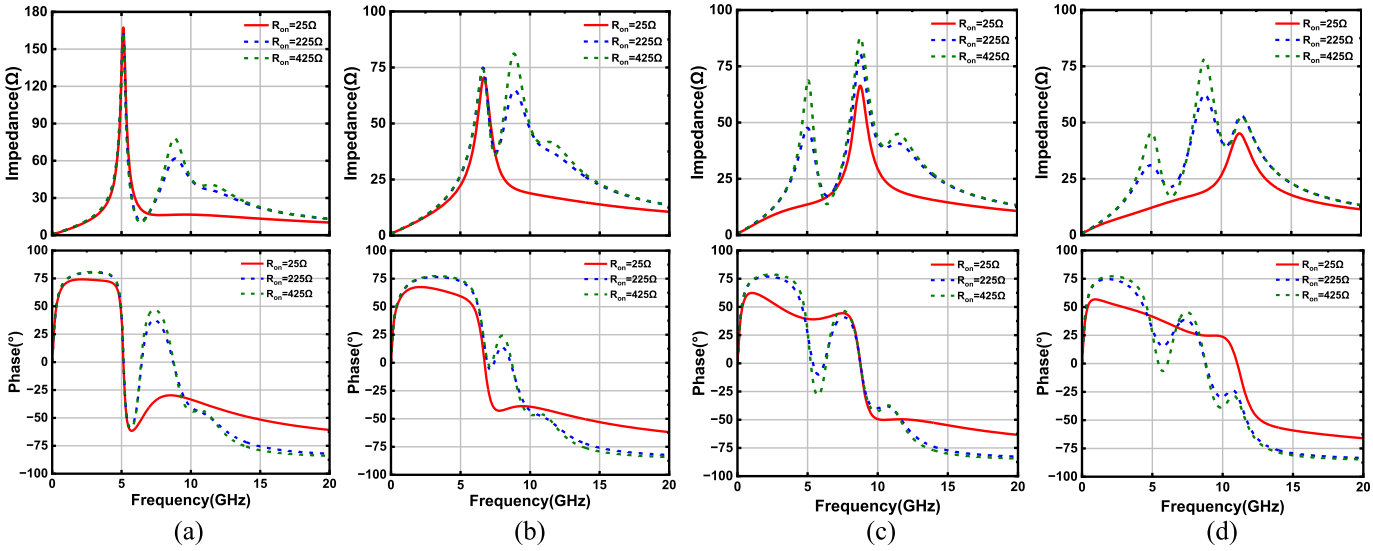


Fig. 14. Simulated magnitude and phase response of the tank impedance under different R_{ON} for (a) mode-1, (b) mode-2, (c) mode-3, and (d) mode-4.

VCOs, an embedded switched inductor (L_{sw}) and a mode-switching (SW) array. The active cores (Core-1, -2, -3, and -4), each composed of cross-coupled transistors, a tail resistor array, a pair of varactors and 6-bit SCA, are identical. The tail resistor array, shown in the bottom-left of Fig. 13(a), consists of a bypass path and a 13-bit programmable resistor bank. It is added to tune the core transconductance G_m versus frequency so that the excess loop gain $G_X = G_m R_T$ (with R_T the tank's equivalent parallel resistance) remains moderate, which in turn limits flicker noise up-conversion [26]. The 6-bit SCA, shown in the top-left of Fig. 13(a), and the MOS varactors provide coarse tuning and continuous fine tuning, respectively. The varactor is implemented using a thick-oxide device, with dimensions of $W/L = 32/0.4$ μm. As the bias increases from 0 to 2.5 V, the varactor capacitance decreases from 35.7 to 15.2 fF ($\Delta C = 20.5$ fF) with Q larger than 29. The SCA exhibits an LSB of 7.7 fF, thereby enabling fine and continuous frequency tuning. With a high C_{max}/C_{min} ratio preserved and Q kept moderate, the total capacitance ranges from 191.9 to 732.1 fF, and Q remains above 24 at 10 GHz. Each core is connected to its respective inductor ports in a center-symmetric manner, ensuring uniform coupling and balanced performance across all modes.

Furthermore, to maintain uniform capacitive loading at each port, the SW array is designed with four identical sub-arrays, all positioned at the center of the transformer symmetrically. Each sub-array contains four switches and there are a total of sixteen switches in all. An additional switch is positioned along the routing of the switched inductor to control its on/off state. As discussed above, eight switches need to be turned on and another eight switches need to be turned off in each mode. The switch on-off states in each mode are shown in Fig. 13(b). As observed, these switches are divided into five groups and are controlled by 5-bit control bits, respectively. In mode-1, Groups-1 and -2 are activated; in mode-2, Groups-1, -2, and -5 are active. Mode-3 enables Group-1 and -3, whereas mode-4 enables Group-3 and -4. Unless otherwise specified, all other switch groups remain disabled.

Given the careful consideration of the symmetry of the inductive, capacitive, active, and other elements in the quad-core quad-mode VCO, the mismatches among VCO cores are significantly minimized, allowing the four single-core VCOs to be well synchronized and achieve optimized PN in all modes.

B. Avoiding Mode Ambiguity

It is known that when the transconductance pairs are sufficiently large, the oscillator operates at the frequency where the tank impedance provides sufficiently high amplitude with zero phase. When no mode switches are added or the switch size is too small [i.e., large ON-resistance (R_{ON})], multiple frequency points with high impedance amplitude and zero phase may exist. As a result, the operating frequency of the oscillator is not uniquely defined, resulting in mode ambiguity.

Due to the symmetrical property of the proposed transformer, the tank impedance seen from node pairs P_1 and P_2, P_3 and P_4, P_5 and P_6, P_7 and P_8 is identical. Fig. 14 shows the simulated magnitude and phase responses of the tank impedance seen from the node pair P_1 and P_2 in the four operation modes, as R_{ON} is reduced from 425 to 25 Ω. When R_{ON} is 225/425 Ω, as shown by the blue/green dotted curve in Fig. 14, each mode exhibits at least two frequency points where the tank impedance has a sufficiently high peak and a zero phase. This implies that the oscillator has the potential to oscillate either at the frequency with the expected maximum Q value or concurrently at multiple resonance frequencies, which are highly undesirable.

To avoid the issues associated with ambiguous modes, it is necessary to reduce R_{ON} by increasing the size of the mode switches. As shown by the solid red curves in Fig. 14, when R_{ON} is reduced to 25 Ω, each mode exhibits a uniquely defined resonant frequency.

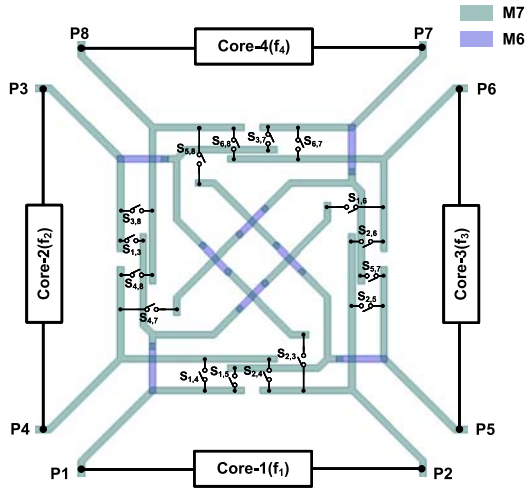


Fig. 15. Center-symmetric layout of mode-switching array.

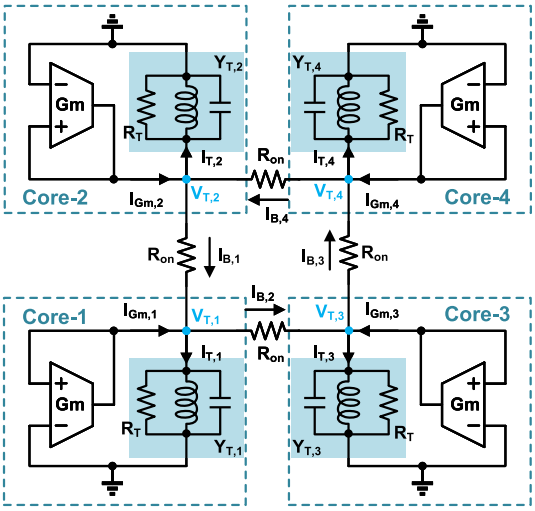


Fig. 16. R_{ON}-coupled equivalent circuit of the proposed VCO.

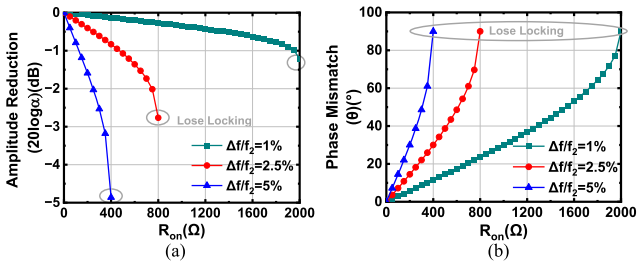


Fig. 17. Calculated (a) amplitude reduction and (b) phase mismatch over R_{ON} under different frequency mismatches with $Q = 15$ and $R_p = 150\ \Omega$.

C. Effect of Mismatches Among Cores

While ideal synchronization assumes identical oscillation frequencies among the cores, practical implementations often suffer from frequency mismatches caused by device variations and layout asymmetries. These mismatches introduce phase errors and amplitude reduction, ultimately degrading the PN performance [21], [27]. Therefore, accurately characterizing and mitigating inter-core mismatches are essential to fully unlock the potential of multi-core VCO designs.

The proposed design makes every effort to ensure symmetry among the four VCO cores, including the transformer tank, SW array, VCO cores and layout routings. As presented in Fig. 9(a), each single tank inductor is identical and arranged in a centrally symmetric layout. Additionally, the SW array, consisting of sixteen switches, is also arranged in a center-symmetric configuration, as illustrated in Fig. 15. That is to say, each port (P_{1-8}) is identically connected to four switches to ensure that the parasitics induced by the switches is the same across all ports. Moreover, each VCO core is connected to the SW array by identical routings, and synchronized with other three cores directly through the SW array, thus further reducing frequency mismatches and mitigating its impact on the VCO performance.

However, due to the presence of the switched inductor, as shown in Fig. 9(a), the effective inductances seen by core-2 and -3 slightly differs from that of core-1 and -4. As a result, the natural oscillation frequencies of core-2 (f_2) and core-3 (f_3) deviate slightly from those of core-1 (f_1) and core-4 (f_4), introducing mismatches in the system. Despite the mismatch, the four oscillator cores may still achieve frequency locking and oscillate at a common frequency located between f_1, f_2, f_3 and f_4. As a result, the cores operate off-resonance, such that $\angle Y_{T,i} \neq 0$, where $Y_{T,i}$ denotes the tank admittance corresponding to core-i, and is given by

$$Y_{T,i} = \frac{1}{R_T} + j\frac{Q}{R_T}\left(\frac{f_{\text{osc}}}{f_i} - \frac{f_i}{f_{\text{osc}}}\right) \cong \frac{1}{R_T} + j\frac{2Q}{R_T}\frac{f_{\text{osc}} - f_i}{f_i} = \frac{1}{R_T} + j\frac{2K_i}{R_T}, \quad i = 1, 2, 3, 4 \tag{24}$$

where R_T is the equivalent parallel resistance of the tank, and $K_i = Q \cdot (f_{\text{osc}} - f_i)/f_i$, with Q, f_{osc}, and f_i representing the quality factor of the tank, the oscillation frequency, and the resonance frequency of core-i, respectively. This approximation holds when $\delta f_i = f_{\text{osc}} - f_i \ll f_i$. Follow the analysis in [27], the proposed VCO is modeled in Fig. 16. Here, R_{ON} denotes the ON-resistance of the mode switch. Assuming equal amplitudes, the tank voltage of each core is written as $V_{T,i} = Ae^{j\theta_i} = \alpha A_0 e^{j\theta_i}$, where A_0 denotes the amplitude in the absence of mismatch, α represents the amplitude reduction factor due to mismatch, and θ_i is the voltage phase corresponding to core-i.

Due to layout symmetry, the switched inductor has an identical effect on core-1 and -4, leading to $Y_{T,1} = Y_{T,4}$, and consequently $\theta_1 = \theta_4$. For simplicity, we assume that $\theta_1 = \theta_4 = 0$ as the reference phase. Through derivation presented in the Appendix, the following results are obtained as:

$$\theta_2 = -\sin^{-1}(K_2 r) \tag{25}$$

$$\alpha = \frac{r + 2\sqrt{1 - (K_2 r)^2} + 2}{4rK_2^2 + r + 4} \tag{26}$$

$$K_2 = \frac{1}{2}Q\frac{f_1 - f_2}{f_2} = -\frac{1}{2}Q\frac{\Delta f}{f_2} \tag{27}$$

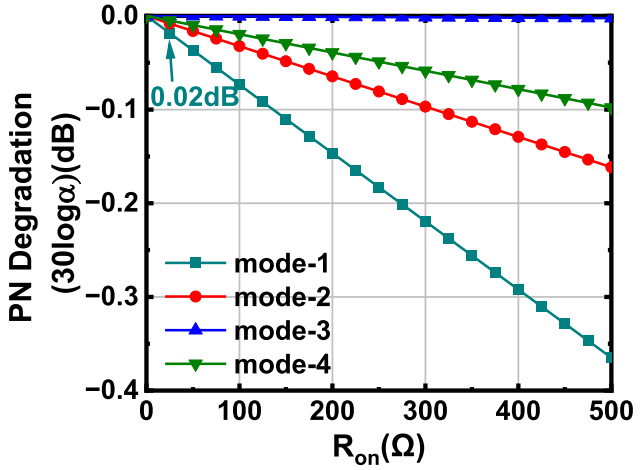


Fig. 18. Calculated PN degradation over R_{ON} in different modes (mode-1: $Q_1 = 16$, $f_1 = 5$ GHz, $L_1 = 1298$ pH; mode-2: $Q_2 = 8.9$, $f_2 = 8$ GHz, $L_2 = 786$ pH; mode-3: $Q_3 = 10.1$, $f_3 = 10$ GHz, $L_3 = 429$ pH; mode-4: $Q_4 = 8.5$, $f_4 = 14$ GHz, $L_4 = 272$ pH).

where $r = R_{ON}/R_T$ and $\Delta f/f_2$ represents the frequency mismatch between core-1 (core-4) and core-2 (core-3).

Based on (25)–(27), the variations of amplitude reduction ($20\log\alpha$) and phase mismatch ($\theta = \theta_2 - \theta_1$) with respect to the frequency mismatch ($\Delta f/f_2$) can be compared as a function of R_{ON}, as shown in Fig. 17. As R_{ON} and the frequency mismatch increase, the amplitude reduction becomes more significant, and the phase deviation between cores also increases. When R_{ON} becomes sufficiently large, the multi-core system can no longer maintain frequency locking, which corresponds to the point where the curves terminate as "lose locking," as shown in Fig. 17.

In the proposed design, the asymmetry caused by the switched inductor introduces frequency mismatches of approximately 2.4%, 2.1%, 0.2%, and 1.3% in mode-1, -2, -3, and -4, respectively, obtained from EM simulations. The PN degradation of each mode is affected by amplitude reduction (α) and phase mismatch (θ), where their combined effect can be approximated as $30\log\alpha$ [27]. Accordingly, Fig. 18 shows the PN degradation versus R_{ON} for the four modes. It is observed that the PN of mode-1 degrades more severely than that of other modes. However, with a relatively small ON-resistance (25 Ω) and a 2.4% frequency mismatch, the PN degradation in mode-1 is only about 0.02 dB, indicating that the asymmetry induced by the switched inductor has a minimal impact on the overall PN performance, thereby effectively approaching full symmetry.

Process variations and some other non-idealities can also introduce mismatches across oscillator cores, which in turn lead to frequency mismatches. To evaluate the impact of process variations and other non-idealities on the PN performance, frequency mismatches are deliberately introduced among the VCO cores by changing the per-core tank capacitance for simulations, covering six cases, as shown in Fig. 19. Fig. 19(a) illustrates a single-core mismatch, where only Core-1 is perturbed by adding 100, 200, or 300 fF; Cores-2/3/4 remain nominal. Fig. 19(b) and (c) evaluate two-core perturbations: (b) Cores-1/2 are perturbed together by adding 100, 200, or 300 fF, while (c) repeats the test of (b) on the opposite pair (Cores-1/4); the other two cores remain nominal. Fig. 19(d) and (e) address mismatches among three frequencies: (d) Cores-1/2 remain nominal while Core-3 has 150 fF (10% mismatch) added and Core-4 is swept over 100, 200, and 300 fF; (e) repeats the test of (d) with the perturbed cores exchanged. In Fig. 19(f), Core-2 remains nominal; Core-1 and Core-3 have 50 fF (4% mismatch) and 150 fF (10% mismatch) added, respectively, while Core-4 is swept over 100, 200, and 300 fF.

Thanks to the well-synchronized properties and low switching resistance R_{ON}, the impact of frequency mismatches on figure-of-merit (FoM) degradation remains minimal. Monte Carlo shows an approximate 9% ($\pm 3\sigma$) frequency mismatch. Therefore, the FoM is evaluated under a conservative 10% mismatch, yielding a penalty of $\leq$1.6 dB in mode-1 and $<$0.5 dB in the other modes.

D. PN Analysis

Having discussed the effect of frequency mismatch on PN, we now examine the influence of R_{ON} on PN in the absence of frequency mismatch. Following the derivation in [29], the PN expression for a symmetrical quad-core VCO is given by

$$\mathcal{L}_{\text{quad-core}}(\Delta\omega) = 10\log\left(F\frac{4kT\left|Z_{\text{eq}}\right|^2}{V_0^2 R_T}\right) \tag{28}$$

where F is the device excess noise factor, T is the absolute temperature, V_0 is the oscillation amplitude and R_T is the equivalent parallel resistance of the tank. $|Z_{\text{eq}}|^2$ can be denoted by

$$\left|Z_{\text{eq}}\right|^2 = \frac{1}{4}\left|Z_T(\Delta\omega)\right|^2\frac{16R_{ON}^2 + \left|Z_T(\Delta\omega)\right|^2}{4R_{ON}^2 + \left|Z_T(\Delta\omega)\right|^2} \tag{29}$$

where $Z_T(\Delta\omega)$ represents the tank impedance and can be given by

$$\left|Z_T(\Delta\omega)\right| = \left|\frac{1}{2}\frac{R_T\omega_0}{Q\cdot\Delta\omega}\right| \tag{30}$$

where ω_0 and $\Delta\omega$ are the oscillation frequency and the offset frequency, respectively, and Q is the equivalent quality factor of the tank. Substituting (29) and (30) into (28) yields

$$\mathcal{L}_{\text{quad-core}}(\Delta\omega) = \mathcal{L}(\Delta\omega)\cdot m. \tag{31}$$

Comparing (31) with (1), it is evident that (31) contains an additional factor m, expressed as follows:

$$m = \frac{1}{4}\frac{16R_{ON}^2 + |Z_T(\Delta\omega)|^2}{4R_{ON}^2 + |Z_T(\Delta\omega)|^2}. \tag{32}$$

It can be observed that both R_{ON} and $\Delta\omega$ have an impact on the PN improvement. When R_{ON} is much smaller than $Z_T(\Delta\omega)$, the value of m is approximately 1/4 and the PN is improved by approximately 6 dB compared to a single-core VCO. On the other hand, in the case when R_{ON} is much larger than $Z_T(\Delta\omega)$ (i.e., $m \approx 1$), the PN approaches that of a single-core VCO, indicating that the four oscillator cores are no longer frequency-locked and instead operate as independent

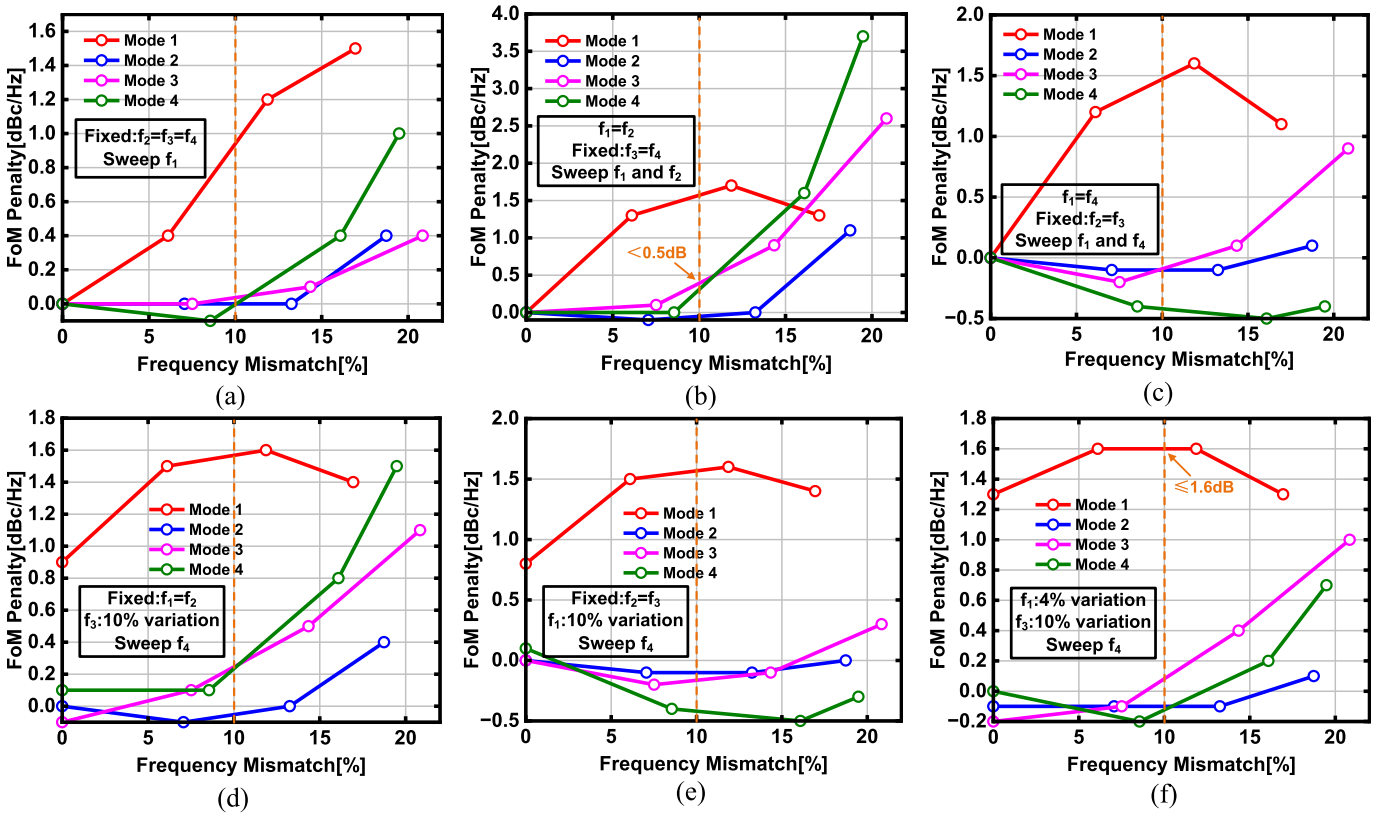


Fig. 19. Impact of frequency mismatches on FoM degradation.

TABLE I

PERFORMANCE SUMMARY AND COMPARISON WITH STATE-OF-THE-ART OSCILLATORS

	This work	L. Chao JSSC'23 [21]	Y. Shu ISSCC'20 [18]	J. Gong ISSCC'22 [15]	O. El-Aassar JSSC'21 [14]	Z. Kang ISSCC'24 [28]	H. Ge ISSCC'24 [22]
Topology	Quad-core Quad-mode	Quad-core Quad-mode	Quad-core Quad-mode	Dual-core Tri-mode	Dual-core Tri-mode	Single core Dual-mode	Quad-core Quad-mode
Technology	65-nm CMOS	65-nm CMOS	40-nm CMOS	22-nm FinFET	22-nm FDSOI	40-nm CMOS	65-nm CMOS
Techniques[#]	M	M	M+E	M+E	M+E	/	M+E
Frequency [GHz]	4.21 to 15.18	12.1 to 32	18.6 to 40.1	7.1 to 16.8	8 to 17	8.9 to 21.9	13.7 to 41.5
TR [%]	**113.2**	90.2	73.2	80.6	72	84.4	101
FTR Overlap[^] [%]	**10.5 to 14.7**	0.6 to 2.5	4.1 to 5.0	7.6 to 9.7	2.2 to 4.4[*]	2.0	3.0 to 5.4
PN@10MHz [dBc/Hz]	-130.7 to -141.4	-122.1 to -126.7	-122.7 to -130.3	-128.7 to -137.9	-134.7 to -143.1	-131.1 to -138.6	-124.4 to -133.2
FoM@10MHz [dBc/Hz]	180.0 to 190.5	177.4 to 181.7	183.0 to 186.3	185.1 to 187.6	180.7 to 191.7	184.8 to 190.5	184.4 to 194.0
FoM_T@10MHz [dBc/Hz]	201.1 to 211.6	194.3 to 200.8	200.3 to 203.6	203.2 to 205.7	197.8 to 208.8	203.3 to 209.0	204.5 to 214.1
FoM_{TA}@10MHz [dBc/Hz]	209.3 to 219.8	205.2 to 211.7	211.3 to 214.6	216.2 to 218.7	201.9 to 212.9	213.3 to 219.0	213.0 to 222.6
$1/f^3$ PN corner [kHz]	35 to 750	/	220 to 620	460 to 1600	3000	200 to 750	280 to 1300
V_{DD} [V]	0.4	0.8	1.1	1.1	0.45	0.45	0.4/0.55
Power [mW]	1.44 to 14.12	5.6 to 22.5	9 to 15	5.5 to 13.5	17 to 33	5.6 to 14.8	2 to 8.8
Core Area [mm^2]	0.152	0.08	0.08	0.05	0.39	0.1	0.14

[*]Estimated from Figures.

[#] M refers to magnetically reconfigurable technique. E refers to electric coupling via added capacitors to change the effective capacitance.

[^]FTR Overlap=$2*(f_{max,mode,n}-f_{min,mode,n+1})/(f_{max,mode,n}+f_{min,mode,n+1})$

FoM=$|PN|+20\log10(f_0/\Delta f)-10\log10(P_{DC}/1mW)$. $FoM_T=FoM+20\log10(FTR[\%]/10)$ $FoM_{TA}=FoM_T+10\log10(1mm^2/Area)$

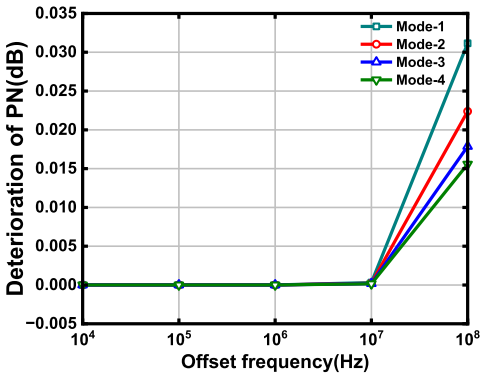


Fig. 20. PN degradation of the proposed VCO for $R_{ON} = 25\ \Omega$, based on (28).

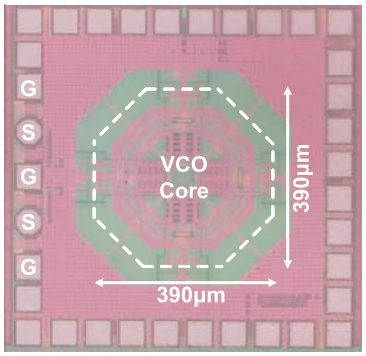


Fig. 21. Chip micrograph of the proposed quad-core quad-mode VCO.

free-running oscillators. Fig. 20 illustrates the calculated PN degradation of the proposed VCO with $R_{ON} = 25\ \Omega$ across all four modes as a function of frequency offsets, based on (28). The PN degradation is nearly zero, thus a near-6 dB PN reduction is still achieved. Relatively speaking, mode-1 exhibits the most severe degradation, but even at a 100 MHz offset, the degradation is less than 0.035 dB, which demonstrates excellent synchronization performance of the proposed quad-core VCO.

IV. Measurement Results

The proposed quad-core quad-mode VCO is implemented in 65 nm bulk CMOS with a core chip area of 0.152 mm^2, as shown in Fig. 21. Fig. 22 shows the measured PN at 4.28, 8.66, 11.57, and 15.07 GHz, respectively, from four distinct modes. The measured FTR is 113.2% (4.21–15.18 GHz), with the frequency in each mode ranging from 4.21 to 6.24 GHz, 5.62 to 8.66 GHz, 7.52 to 11.6 GHz, and 10.01 to 15.18 GHz, respectively, as shown in Fig. 23. In each mode, the tuning voltage ranges from 0 to 2.5 V. It's worth noting that the minimum frequency overlap between two adjacent modes is 10.5%, which further validates the robustness of the wide bandwidth. We further report the VCO gain K_{VCO} on this plot at $V_{tune} = 1.2\ V$. Fig. 24(a) and (b) show the measured PN, FoM and FoM$_T$ over the whole FTR, respectively, where the PN is from -108.8 to -120.3 dBc/Hz and -130.7 to -141.4 dBc/Hz at 1 and 10 MHz offsets while the FoM and FoM$_T$ vary from 180.0 to 190.5 dBc/Hz and 201.1 to 211.6 dBc/Hz at a 10 MHz offset.

The tank's effective parallel resistance R_T varies by mode, owing to variations in Q across modes. Since mode-1 exhibits the largest Q and thus the largest R_T, it requires the smallest g_m and hence the lowest current consumption, while the higher frequency and lower Q modes require more. In the proposed design, a programmable tail resistor array is used to adjust the current. In mode-1, a larger tail resistor is used to moderate g_m, while smaller resistors are employed in modes-2 and -4 to allow higher g_m. The VCO consumes 1.44 to 14.12 mW power under a VDD of 0.4 V, with mode-1: 1.44–3.27 mW; mode-2: 9.51–13.46 mW; mode-3: 11.14–14.10 mW; and mode-4: 12.63–14.12 mW. Although the 0.4 V low supply voltage helps reduce intrinsic power consumption of the VCO, an on-chip regulator is typically required to suppress noise and improve the power supply rejection ratio (PSRR), which results in additional power overhead. Across different modes, the K_{push} of the proposed VCO is measured to range from 126 to 297 MHz/V. Since spur level depends on the PSRR, an on-chip regulator with high PSRR is required to keep spurs low. Although magnetic coupling brings Q variations, the tail resistor array ensures the robust start-up and also suppresses flicker noise up-conversion across a wide FTR. This effect is reflected in the 1/f^3 PN corners in each mode, which span 35–220 kHz (mode-1), 130–210 kHz (mode-2), 180–600 kHz (mode-3), and 90–750 kHz (mode-4). A lower corner frequency indicates more effective suppression of flicker noise up-conversion. Thanks to the tail resistor, the normalized PN (to 1 GHz) shows good flatness with minimal variations across the whole FTR, as shown in Fig. 24(a). Moreover, Fig. 25 shows that the proposed VCO exhibits the best overall performance in view of FoM and FTR, and the best FoM$_{TA}$ in similar operating frequencies. The results are compared with state-of-the-art wideband VCOs in Table I, where the proposed VCO exhibits the largest FTR and the largest FTR overlap facilitated by MR technique, validating the robustness of the ultra-wide bandwidth.

V. Conclusion

In this article, we present a quad-core quad-mode VCO featuring MR technique and fully symmetrical topology. By changing the current directions of the tank inductors via the mode-switching array and incorporating the switched inductor, four distinct inductance values are realized, enabling quad-mode operation without electrical coupling. Besides, with careful consideration of the symmetry in the inductive, capacitive, and active elements, as well as the routing layout, mismatches among the VCO cores are significantly minimized, resulting in optimized PN performance. The VCO achieves an FTR of 113.2% (4.21-to-15.18 GHz), a peak FoM$_T$ of 211.6 dBc/Hz and a peak FoM$_{TA}$ of 220.5 dBc/Hz at a 10 MHz offset.

Appendix

In Fig. 16, the current relations at the blue-highlighted nodes satisfy Kirchhoff's current law (KCL), as expressed by

$$I_{Gm,1} + I_{B,1} - I_{B,2} = I_{T,1} \tag{33}$$

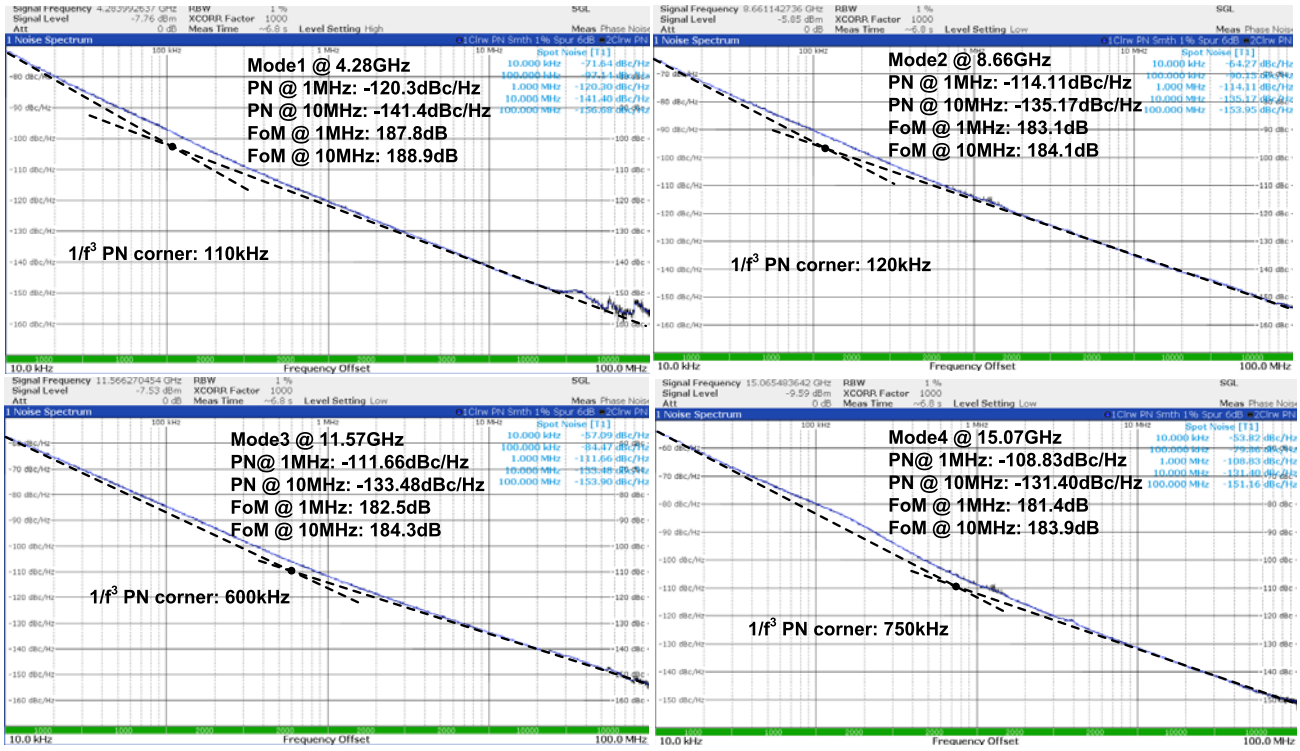


Fig. 22. Measured PN performance of each mode.

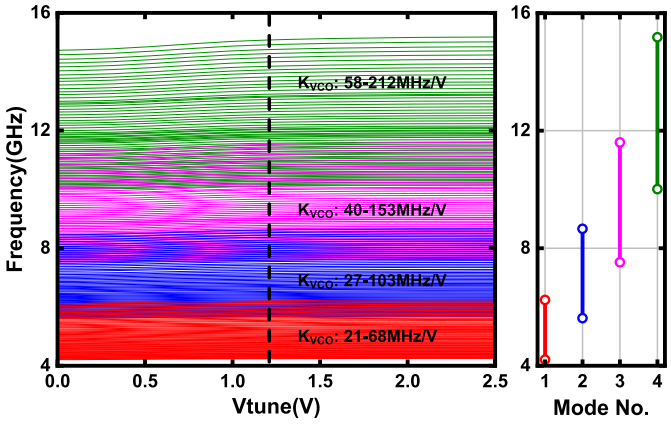


Fig. 23. Measured FTR in four modes.

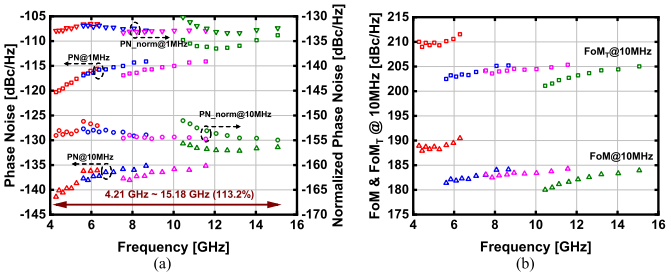


Fig. 24. Measured (a) PN over the FTR and (b) FoM and FoM$_T$ over the FTR.

$$I_{Gm,2} + I_{B,4} - I_{B,1} = I_{T,2} \tag{34}$$

$$I_{Gm,3} + I_{B,2} - I_{B,3} = I_{T,3} \tag{35}$$

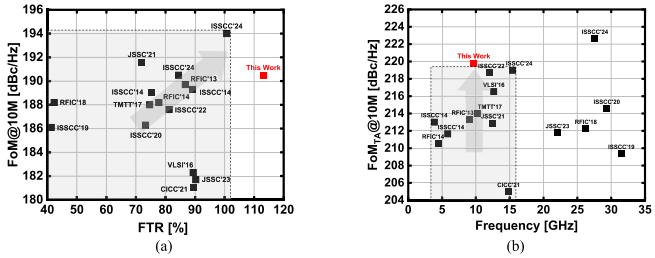


Fig. 25. Comparison of (a) FoM over the FTR and (b) FoM$_{TA}$ over the frequency at a 10 MHz offset with recently reported wideband VCOs.

$$I_{Gm,4} + I_{B,3} - I_{B,4} = I_{T,4} \tag{36}$$

where $I_{B,1} = (V_{T,2} - V_{T,1})/R_{\text{on}}$ and $I_{T,1} = V_{T,1}Y_{T,1}$. The other parameters can be derived in similar manner.

For simplicity, we assume that $|I_{Gm,1}| = |I_{Gm,2}| = |I_{Gm,3}| = |I_{Gm,4}| = 1$, which accordingly implies that $A_0 = R_T$. Moreover, $I_{Gm,i}$ is in phase with $V_{T,i}$. Substituting the parameters ($I_{Gm,2}$, $I_{B,4}$, $I_{B,1}$, and $I_{T,2}$) into (34), we obtain

$$e^{j\theta_2} + \frac{\alpha}{r}\left(2 - 2e^{j\theta_2}\right) = \alpha\left(e^{j\theta_2}\right)(1 + j2K_2) \tag{37}$$

where $r = R_{\text{ON}}/R_T$. Equating the real and imaginary parts on both sides of (37), we obtain

$$\frac{2\alpha}{r}(1 - \cos\theta_2) + \cos\theta_2 = \alpha\cos\theta_2 - 2\alpha K_2 \sin\theta_2 \tag{38}$$

$$\left(1 - \frac{2\alpha}{r}\right)\sin\theta_2 = \alpha\sin\theta_2 + 2\alpha K_2\cos\theta_2. \tag{39}$$

Solving (38) and (39) yields θ_2 and α as given in (25) and (26).

Using the same approach to calculate (35) and comparing the corresponding results of θ_3 and α with (25) and (26), we obtain $K_2 = K_3$ and $\theta_2 = \theta_3$. The above solutions are valid when the condition $|K_2 r| \leq 1$ is satisfied.

Similarly, solving the imaginary part of (33), we obtain

$$\frac{\alpha}{r}(\sin\theta_2 + \sin\theta_3) = 2\alpha K_1. \tag{40}$$

Substituting (25) into (40), it can be derived that

$$-(K_2 + K_3) = 2K_1. \tag{41}$$

Based on (41), and noting that $K_1 = K_4$ and $K_2 = K_3$, it can be approximated that $f_{\text{osc}} \approx (f_1 + f_2 + f_3 + f_4)/4$, which is consistent with the result in [27]. Accordingly, by substituting f_{osc} into K_2, it can be rewritten as (27).

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